

CLMS Migration to Copernicus Data Space Ecosystem

Status Update



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```

// — Load data from the canonical data repo —
data = await fetch("https://raw.githubusercontent.com/copernicus-land/clms-cdse-migration-data/main/migration_data.json")
  .then(r => { if (!r.ok) throw new Error(String(r.status)); return r.json() })
  .catch(e => { console.error("Failed to load data", e); return {groups:[], portfolio:{}, cdse:{}, summary:{}} })
groups = data.groups
summary = data.summary
portfolio = data.portfolio
cdse = data.cdse
lastUpdated = data.scraped_at.slice(0,10)

// — OData tracking lives in the main data file (single JSON) —
odataDaily = data.odata_tracking || {}

// — OData deltas, computed client-side from the daily totals history —
odataHist = odataDaily.history || []
odataTotal = odataDaily.total_odata || (odataHist.length ? odataHist[odataHist.length - 1].total : 0)
odataDailyDelta = odataHist.length ≥ 2
  ? odataHist[odataHist.length - 1].total - odataHist[odataHist.length - 2].total
  : 0
odataWeeklyBase = {
  if (odataHist.length === 0) return null
  const today = odataHist[odataHist.length - 1].date
  const weekAgo = new Date(new Date(today).getTime() - 7 * 86400000).toISOString().slice(0, 10)
  const past = odataHist.filter(h => h.date ≤ weekAgo)
  return past.length ? past[past.length - 1] : odataHist[0]
}
odataWeeklyDelta = odataWeeklyBase ? odataTotal - odataWeeklyBase.total : 0
odataDaysCovered = odataWeeklyBase && odataHist.length
  ? Math.max(1, Math.round((new Date(odataHist[odataHist.length - 1].date) - new Date(odataWeeklyBase.date)) / 86400000))
  : 1
odataDailyAvg = odataWeeklyDelta ? Math.round(odataWeeklyDelta / odataDaysCovered) : 0

// — Filter groups —
componentOf = d => {
  if (d.datasets?.length > 0) return d.datasets[0].component
  return d.portfolio?.[0]?.component || ""
}
groupHasComponent = (g, comp) => {
  if (g.datasets?.some(d => d.component === comp)) return true
  if (g.portfolio?.some(d => d.component === comp)) return true
  return false
}
filtered = groups.filter(d => {
  let matches = false
  for (const comp of activeComponents) {
    if (groupHasComponent(d, comp)) { matches = true; break }
  }
  if (!matches) return false
  if (searchTerm && !d.name.toLowerCase().includes(searchTerm.toLowerCase())) return false
  return true
})

```

```
// — Aggregate by category —
categories = [...new Set(filtered.map(d => d.category).filter(Boolean))].sort()

// — Totals reflecting the current filter selection —
filteredTotals = ({
  datasets: d3.sum(filtered, d => d.total),
  groups: filtered.length
})
catStats = categories.map(cat => {
  const g = filtered.filter(d => d.category === cat)
  return {
    name: cat,
    groups: g,
    total: d3.sum(g, d => d.total),
    on_cdse: d3.sum(g, d => d.on_cdse),
    pct: Math.round(d3.sum(g, d => d.on_cdse) / Math.max(d3.sum(g, d => d.total), 1) * 100),
    s3: d3.sum(g, d => d.s3),
    stac: d3.sum(g, d => d.stac),
    openeo: d3.sum(g, d => d.openeo),
    shub: d3.sum(g, d => d.shub),
  }
})

// — Helpers —
badge = (val, total) => {
  if (val === 0 && total === 0) return htl.html`<span style="background:#888;color:#fff;padding:1px 6px;border-radius:3px;font-weight:bold"></span>`
  const pct = Math.min(val / Math.max(total, 1), 1)
  const color = pct ≥ 1 ? "#00aa00" : pct ≥ 0.75 ? "#66cc00" : pct ≥ 0.5 ? "#ffcc00" : pct ≥ 0.25 ? "#ff8800" : "#ff0000"
  return htl.html`<span style="background:${color};color:#fff;padding:1px 6px;border-radius:3px;font-weight:bold">${val}</span>`
}

check = (on) => on
  ? htl.html`<span style="color:#00aa00;font-weight:bold">✓</span>`
  : htl.html`<span style="color:#ccc">×</span>`

// — Deltas vs previous run, computed client-side from the embedded snapshot —
prevByName = Object.fromEntries((data.previous?.groups || []).map(g => [g.name, g]))
changes = (data.previous?.groups?.length ? groups : []).map(g => {
  const old = prevByName[g.name] || {}
  const oldIds = new Set(old.cdse_dataset_ids || [])
  const oldStac = new Set(old.stac_ids || [])
  const oldOeo = new Set(old.openeo_ids || [])
  const curIds = (g.datasets || []).map(d => d.id)
  const dS3 = g.s3 - (old.s3 || 0)
  const dStac = g.stac - (old.stac || 0)
  const dOeo = g.openeo - (old.openeo || 0)
  return {
    name: g.name,
    s3: dS3, stac: dStac, openeo: dOeo,
    newly_on_cdse: g.on_cdse > 0 && !(old.on_cdse > 0),
    new_s3: dS3 > 0 ? curIds.filter(id => !oldIds.has(id)) : [],
  }
})
```

```
new_stac: dStac > 0 ? curIds.filter(id => !oldStac.has(id)) : [],
new_openeo: d0eo > 0 ? curIds.filter(id => !old0eo.has(id)) : [],
}
}).filter(d => d.s3 !== 0 || d.stac !== 0 || d.openeo !== 0)
newGroups = changes.filter(d => d.newly_on_cdse)
positiveChanges = changes.filter(d => d.s3 > 0 && !d.newly_on_cdse)
lastWeek = changes.length > 0 ? "since " + (data.previous?.scraped_at || data.scraped_at).slice(0,10) : ""

// — Expand/collapse state, kept outside the DOM so it survives the full
// table rebuild that happens on every filter or search change —
expandedCategories = new Set()
expandedGroups = new Set()

// — Toggle functions for expand/collapse (plain DOM, no OJS reactivity) —
toggleScript = {
  window.toggleCategory = function(catName) {
    const key = catName.replace(/"/g, '"');
    const groupRows = document.querySelectorAll('tr[data-category="' + key + '"]:not([data-group])');
    const isExpanded = groupRows.length > 0 && groupRows[0].style.display !== 'none';
    groupRows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    if (isExpanded) {
      // Collapsing the category also collapses any open dataset rows within it
      document.querySelectorAll('tr[data-category="' + key + '"][data-group]').forEach(function(r) { r.style.display = 'none'; });
      document.querySelectorAll('span[data-group-arrow="' + key + '"]').forEach(function(a) { a.textContent = '▶'; });
      expandedCategories.delete(catName);
      [...expandedGroups].filter(k => k.startsWith(catName + '::')).forEach(k => expandedGroups.delete(k));
    } else {
      expandedCategories.add(catName);
    }
    const arrow = document.querySelector('span[data-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
  window.toggleGroup = function(groupKey) {
    const key = groupKey.replace(/"/g, '"');
    const rows = document.querySelectorAll('tr[data-group="' + key + '"]');
    const isExpanded = rows.length > 0 && rows[0].style.display !== 'none';
    rows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    if (isExpanded) expandedGroups.delete(groupKey);
    else expandedGroups.add(groupKey);
    const arrow = document.querySelector('span[data-group-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
}
```

```
// — Onboarding history plot —
```

```
onboardingPlot = {
  const barHeight = 28
  let mode = "daily"
```

```
function getWeekNumber(dateStr) {
  const d = new Date(dateStr)
  const start = new Date(d.getFullYear(), 0, 1)
  const days = Math.floor((d - start) / 86400000)
  return Math.ceil((days + start.getDay() + 1) / 7)
}

function buildChart(container) {
  container.innerHTML = ""

  const data = mode === "daily"
    ? odataHist.map((h, i) => {
        const d = i > 0 ? h.total - odataHist[i - 1].total : 0
        return {label: h.date, value: d, color: d ≥ 0 ? '#e67e22' : '#e74c3c'}
      }).slice(1).reverse()
    : [{label: "WoY " + getWeekNumber(odataDaily.scraped_at || ""), value: odataWeeklyDelta,
      color: '#e67e22'}]

  const maxVal = Math.max(...data.map(d => Math.abs(d.value)), 1)

  data.forEach(d => {
    const pct = Math.abs(d.value) / maxVal
    const row = document.createElement("div")
    row.style.cssText = "display:flex;align-items:center;margin:2px 0;font-size:0.85em"

    const label = document.createElement("span")
    label.textContent = d.label
    label.style.cssText = "width:100px;text-align:right;padding-right:8px;color:#555;flex-shrink:0"
    row.appendChild(label)

    const track = document.createElement("div")
    track.style.cssText = `height:${barHeight}px;flex:1;display:flex;align-items:center`

    const fill = document.createElement("div")
    fill.style.cssText = `height:${barHeight - 4}px;width:${Math.max(pct * 100, 2)}%;background:${d.color};border-radius:3px;display:flex;align-items:center;padding-left:6px;min-width:fit-content`
    fill.textContent = (d.value ≥ 0 ? "+" : "") + Math.abs(d.value).toLocaleString()
    fill.style.color = "#fff"
    fill.style.fontSize = "0.85em"
    fill.style.fontWeight = "bold"
    track.appendChild(fill)
    row.appendChild(track)
    container.appendChild(row)
  })

  // Radio buttons below the chart
  const controls = document.createElement("div")
  controls.style.cssText = "display:flex;gap:16px;margin-top:8px;align-items:center"

  ["daily", "weekly"].forEach(m => {
    const lbl = document.createElement("label")
    lbl.style.cssText = "display:flex;align-items:center;gap:4px;font-
```

```
size:0.85em;cursor:pointer;color:#555"

const radio = document.createElement("input")
radio.type = "radio"
radio.name = "onboardingMode"
radio.value = m
radio.checked = m === mode
radio.style.cursor = "pointer"

radio.addEventListener("change", () => {
  mode = m
  buildChart(container)
})

lbl.appendChild(radio)
lbl.append(m)
controls.appendChild(lbl)
})

container.appendChild(controls)
}

const container = document.createElement("div")
container.style.marginTop = "8px"
buildChart(container)
return container
}
```

1. Migration Statistics

```
html`
${newGroups.length > 0 ? htl.html `
<div style="background:#e8f5e9;border:1px solid #a5d6a7;border-
radius:6px;padding:12px;margin:12px 0">
  <strong>📦 Products ${lastWeek}</strong>
  <div style="margin:4px 0"><strong>Newly onboarded:</strong> ${newGroups.map(d => d.name).join(",
")}</div>
</div>` : ""}

${positiveChanges.length > 0 ? htl.html `
<div style="background:#e8f5e9;border:1px solid #a5d6a7;border-radius:6px;padding:12px 12px 4px
12px;margin:12px 0">
  <strong>📦 Products ${lastWeek}</strong>
  ${positiveChanges.map(d => htl.html `<div style="margin:4px 0">${d.name}: <strong>+${d.s3}</
strong> S3, <strong>+${d.stac}</strong> STAC, <strong>+${d.openeo}</strong> OpenEO</div>`)}
</div>` : ""}

${changes.length === 0 ? htl.html `<div style="background:#f5f5f5;border:1px solid #ddd;border-
radius:6px;padding:12px;margin:12px 0;color:#888">No new CLMS product were onboarded this week.</
div>` : ""}

${(odataWeeklyDelta || odataDailyDelta || odataHist.length > 0) ? htl.html `
```

```
<div style="background:#fff3e0;border:1px solid #ffcc80;border-radius:6px;padding:12px;margin:12px 0">
  <strong>📁 File Level Statistics</strong>
  ${odataDailyAvg ? html.html `<div style="margin:4px 0;color:#888;font-size:0.85em">Daily average:
  ${odataDailyAvg.toLocaleString()} new files/day</div>` : ""}
  ${onboardingPlot}
</div>` : ""}
`
```

2. CLMS Product on CDSE

This table shows the full CLMS portfolio with their migration status to the CDSE.

```
// Toggle-button style filter: click a component to enable/disable it
// (multi-select, all enabled by default). Replaces the old single-select
// dropdown, which only let one component be viewed at a time.
viewof activeComponents = {
  const allComponents = [...new Set(groups.map(d => componentOf(d)).filter(Boolean))].sort()
  const active = new Set(allComponents)

  const container = document.createElement("div")
  container.style.cssText = "display:flex;gap:8px;flex-wrap:wrap;align-items:center;margin-bottom:12px";

  function render() {
    container.innerHTML = "";
    allComponents.forEach(comp => {
      const isActive = active.has(comp);
      const btn = document.createElement("button");
      btn.type = "button";
      btn.textContent = comp;
      btn.style.cssText = `padding:6px 14px;border-radius:16px;cursor:pointer;font-size:0.85em;border:1px solid ${isActive ? "#0d6efd" : "#ccc"};background:${isActive ? "#0d6efd" : "#f5f5f5"};color:${isActive ? "#fff" : "#555"}`;
      btn.addEventListener("click", () => {
        if (active.has(comp)) active.delete(comp);
        else active.add(comp);
        render();
        container.value = new Set(active);
        container.dispatchEvent(new CustomEvent("input"));
      });
      container.appendChild(btn);
    });
  }
  render();
  container.value = new Set(active);
  return container;
}

viewof searchTerm = Inputs.text({label: "Search", placeholder: "Product group...", width: 250})
```



```
// — Build the table via plain DOM APIs —
// so that column widths defined on <colgroup> are guaranteed to apply to
// every row, including rows toggled between hidden/visible.
productGroupsTable = {
  // Category takes the remaining width (~46%) now that Progress is gone
  const colWidths = ["auto", "9%", "9%", "9%", "9%", "9%", "9%"];

  function makeCell(tag, html, opts = {}) {
    const el = document.createElement(tag);
    if (html instanceof Node) el.appendChild(html);
    else el.innerHTML = html;
    el.style.padding = opts.padding ?? "6px";
    el.style.whiteSpace = "nowrap";
    el.style.overflow = "hidden";
    el.style.textOverflow = "ellipsis";
    if (opts.center) el.style.textAlign = "center";
    if (opts.bold) el.style.fontWeight = "bold";
    if (opts.title) el.title = opts.title;
    if (opts.style) el.style.cssText += ";" + opts.style;
    return el;
  }

  const wrap = document.createElement("div");

  const table = document.createElement("table");
  table.style.cssText = "width:100%;border-collapse:collapse;font-size:0.8em;table-layout:fixed";

  const colgroup = document.createElement("colgroup");
  colWidths.forEach(w => {
    const col = document.createElement("col");
    col.style.width = w;
    colgroup.appendChild(col);
  });
  table.appendChild(colgroup);

  const thead = document.createElement("thead");
  const headRow = document.createElement("tr");
  headRow.style.cssText = "background:#f0f0f0;text-align:left";
  ["Category", "Total", "CDSE", "S3", "STAC", "OpenEO", "S.Hub"].forEach((label, i) => {
    const th = makeCell("th", label, { center: i > 0, padding: "6px" });
    th.style.borderBottom = "2px solid #ddd";
    headRow.appendChild(th);
  });
  thead.appendChild(headRow);
  table.appendChild(thead);

  const tbody = document.createElement("tbody");

  catStats.forEach(cat => {
    const tr = document.createElement("tr");
    tr.style.cssText = "background:#f5f5f5;border-bottom:1px solid #ccc;cursor:pointer";

    const nameCell = makeCell("td", "", { bold: true, title: cat.name });
    const arrow = document.createElement("span");
    arrow.dataset.arrow = cat.name;
```

```
arrow.textContent = expandedCategories.has(cat.name) ? "▼" : "►";
nameCell.appendChild(arrow);
nameCell.append(" " + cat.name);
tr.appendChild(nameCell);

tr.appendChild(makeCell("td", String(cat.total), { center: true, bold: true }));
tr.appendChild(makeCell("td", String(cat.on_cdse), { center: true, bold: true }));
tr.appendChild(makeCell("td", badge(cat.s3, cat.total), { center: true }));
tr.appendChild(makeCell("td", badge(cat.stac, cat.total), { center: true }));
tr.appendChild(makeCell("td", badge(cat.openeo, cat.total), { center: true }));
tr.appendChild(makeCell("td", badge(cat.shub, cat.total), { center: true }));

tr.addEventListener("click", () => window.toggleCategory(cat.name));
tbody.appendChild(tr);

cat.groups.forEach(g => {
  const groupKey = cat.name + "::" + g.name;
  const gtr = document.createElement("tr");
  gtr.dataset.category = cat.name;
  gtr.style.cssText = "border-bottom:1px solid #ddd;cursor:pointer" +
    (expandedCategories.has(cat.name) ? "" : ";display:none");

  const gNameCell = makeCell("td", "", { title: g.name, padding: "4px 6px 4px 24px" });
  const dsInFilter = (g.datasets || []).filter(ds => activeComponents.has(ds.component));
  const lp0ffcDse = (g.portfolio || []).filter(lp => !lp.on_cdse &&
    activeComponents.has(lp.component));
  const hasDatasets = dsInFilter.length + lp0ffcDse.length > 0;
  if (hasDatasets) {
    const gArrow = document.createElement("span");
    gArrow.dataset.groupArrow = groupKey;
    gArrow.textContent = expandedGroups.has(groupKey) ? "▼" : "►";
    gArrow.style.marginRight = "4px";
    gNameCell.appendChild(gArrow);
  }
  const link = document.createElement("a");
  link.href = g.url;
  link.target = "_blank";
  link.textContent = g.name;
  link.addEventListener("click", (e) => e.stopPropagation());
  gNameCell.appendChild(link);
  gtr.appendChild(gNameCell);

  gtr.appendChild(makeCell("td", String(g.total), { center: true, bold: true, padding: "4px 6px" }));
  gtr.appendChild(makeCell("td", String(g.on_cdse), { center: true, bold: true, padding: "4px 6px" }));
  gtr.appendChild(makeCell("td", badge(g.s3, g.total), { center: true, padding: "4px 6px" }));
  gtr.appendChild(makeCell("td", badge(g.stac, g.total), { center: true, padding: "4px 6px" }));
  gtr.appendChild(makeCell("td", badge(g.openeo, g.total), { center: true, padding: "4px 6px" }));
  gtr.appendChild(makeCell("td", badge(g.shub, g.total), { center: true, padding: "4px 6px" }));

  if (hasDatasets) {
    gtr.addEventListener("click", () => window.toggleGroup(groupKey));
  }
});
```

```
tbody.appendChild(gtr);

dsInFilter.forEach(ds => {
  const dtr = document.createElement("tr");
  dtr.dataset.category = cat.name;
  dtr.dataset.group = groupKey;
  dtr.style.cssText = "border-bottom:1px solid #eee;background:#fafafa" +
    (expandedGroups.has(groupKey) ? "" : ";display:none");

  dtr.appendChild(makeCell("td", ds.id, { padding: "3px 6px 3px 44px", title: ds.id }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.s3), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.stac), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.openeo), { center: true, padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", check(ds.shub), { center: true, padding: "3px 6px" }));

  tbody.appendChild(dtr);
});

lpOffCdse.forEach(lp => {
  const dtr = document.createElement("tr");
  dtr.dataset.category = cat.name;
  dtr.dataset.group = groupKey;
  dtr.style.cssText = "border-bottom:1px solid #eee;background:#fafafa" +
    (expandedGroups.has(groupKey) ? "" : ";display:none");

  dtr.appendChild(makeCell("td", lp.title, { padding: "3px 6px 3px 44px", title: lp.title,
    style: "color:#999;font-style:italic" }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
  dtr.appendChild(makeCell("td", "-", { center: true, padding: "3px 6px", style:
    "color:#ccc" }));
  dtr.appendChild(makeCell("td", "-", { center: true, padding: "3px 6px", style:
    "color:#ccc" }));
  dtr.appendChild(makeCell("td", "-", { center: true, padding: "3px 6px", style:
    "color:#ccc" }));
  dtr.appendChild(makeCell("td", "-", { center: true, padding: "3px 6px", style:
    "color:#ccc" }));

  tbody.appendChild(dtr);
});

// — Summary row (col sums over the currently filtered groups) —
const sumRow = document.createElement("tr");
sumRow.style.cssText = "background:#e0e0e0;border-top:2px solid #999;font-weight:bold";
const sumName = makeCell("td", "Total", { bold: true });
sumName.style.fontStyle = "italic";
sumRow.appendChild(sumName);

const sumTotal = d3.sum(catStats, d => d.total);
const sumCDSE = d3.sum(catStats, d => d.on_cdse);
const sumS3 = d3.sum(catStats, d => d.s3);
```

```
const sumSTAC = d3.sum(catStats, d => d.stac);
const sumOEO = d3.sum(catStats, d => d.openeo);
const sumSHub = d3.sum(catStats, d => d.shub);

sumRow.appendChild(makeCell("td", String(sumTotal), { center: true, bold: true }));
sumRow.appendChild(makeCell("td", String(sumCDSE), { center: true, bold: true }));
sumRow.appendChild(makeCell("td", badge(sumS3, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumSTAC, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumOEO, sumTotal), { center: true }));
sumRow.appendChild(makeCell("td", badge(sumSHub, sumTotal), { center: true }));
tbody.appendChild(sumRow);

table.appendChild(tbody);
wrap.appendChild(table);
return wrap;
}
```

3. Services

- [S3 CSV Catalogue](#)
- [OData API](#)
- [STAC Browser](#)
- [OpenEO](#)
- [Sentinel Hub](#)
- [CDSE Documentation](#)

4. On CDSE soon

Details to be provided – placeholder for upcoming CLMS products planned for CDSE ingestion.

Preparing dataset descriptions, expected timelines, and service coverage for the next batch of products to be onboarded.

5. Methodology

Per-service counts are matched by:

- **S3:** Direct catalogue count per product type
- **STAC:** Fuzz-matched by checking if CDSE dataset ID (minus `_cog/_nc` suffix) appears in STAC collection IDs
- **OpenEO:** Matched by constructing `CLMS_{id.upper()}` convention
- **Sentinel Hub:** Estimated from S3 availability